

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Computer Network

COURSE CODE: CSA0714

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I Sri Janani B(192421036) (Name / Reg No) certify that this submission is my original work and

that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sri Janani B

Annexure A
Short Answer Test

1. A company needs to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwidth of 50 Mbps. The Transport Layer divides the file into 1500-byte segments, while the Data Link Layer encapsulates each segment into frames by adding 40 bytes of header and trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

Answer:

Given

Parameter	Value
File size	150 MB = 150×10^6 bytes
Bandwidth	50 Mbps
Segment size	1500 bytes
Frame overhead	40 bytes

Number of segments=100,000

Calculation: $150,000,000 / 1500 = 100,000$

Number of frames=100,000

One segment per frame.

Total Data Link overhead

4,000,000 bytes

Calculation: $100,000 \times 40 = 4,000,000$ bytes = 4 MB

Total transmitted data=154 MB

$150 \text{ MB} + 4 \text{ MB} = 154 \text{ MB}$

Transmission time=24.64 seconds

Total bits = $154 \times 10^6 \times 8 = 1.232 \times 10^9$ bits

Time = $1.232 \times 10^9 / 50 \times 10^6 = 24.64 \text{ s}$

Answer

- Segments = 100,000
 - Frames = 100,000
 - Data Link overhead = 4 MB
 - Transmission time $\approx 24.64 \text{ s}$
- Reliability
- Transport Layer: sequencing, acknowledgements, retransmission, flow control.
 - Data Link Layer: framing, error detection (CRC), local retransmission.

2.A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Answer:

Window size=6 frames

Total frames=48

Windows required=8

Calculation: $48 / 6 = 8$

Retransmissions=8

One lost frame per window $\rightarrow 8 \times 1 = 8$

Answer

- Transmission windows = 8
- Total retransmissions = 8 frames

Flow Control

Flow control prevents the sender from overwhelming the receiver, reduces buffer overflow, and keeps the link efficiently **utilized**.

3.A 12,000-byte IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a 20- byte IP header, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

Answer:

Given

Parameter	Value
Packet size	12,000 bytes
MTU	1,500 bytes
IP header per fragment	20 bytes
Payload per fragment	$1,500 - 20 = 1,480$ bytes

Number of fragments

$$12,000 / 1,480 = 8.1 \rightarrow 9 \text{ fragments}$$

Header overhead

$$9 \times 20 = 180 \text{ bytes}$$

Answer

- Fragments created = 9
- Total IP header overhead = 180 bytes

Network Layer Function

The Network Layer adds fragment offsets and identification fields so that the destination can correctly reassemble all fragments.

4. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Answer:

- Windows required = 8
- Total retransmissions = 8 frames

Flow control: balances sender and receiver speeds, improving throughput and reducing congestion.

5. A multimedia application transfers a 600 MB video file. The Session Layer inserts a synchronization checkpoint after every 60 MB of transmitted data. During transmission, a failure occurs after 390 MB has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

Answer:

Given

Parameter	Value
Video size	600 MB
Checkpoint interval	60 MB
Failure after	390 MB
Checkpoints before failure	

$390 / 60 = 6.5 \rightarrow$ checkpoints at 60, 120, 180, 240, 300, 360 MB

Total checkpoints = 6

Data to retransmit

$390 - 360 = 30$ MB

Answer

- Checkpoints created = 6
- Data retransmitted = 30 MB

Importance

Synchronization checkpoints allow communication to resume from the last saved point instead of restarting the entire transfer.

6. Before transmission, the Presentation Layer compresses a 900 MB file by 40%. The compressed data is encrypted, adding 5% overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

Answer:

Given

Parameter	Value
Original file	900 MB
Compression	40%

Encryption overhead 5%

Compressed size=540 MB

$900 \times 0.60 = 540$ MB

Encrypted size=567 MB

$540 \times 1.05 = 567$ MB

Reduction=37%

Reduction = $900 - 567 = 333$ MB

Percentage reduction = $(333 / 900) \times 100 = 37\%$

Answer

- Compressed file = 540 MB
- Encrypted file = 567 MB
- Total reduction = 37%

Presentation Layer Role

- Compression: reduces transmission time and bandwidth usage.
- Encryption: provides confidentiality and data security.

7. An FTP application transfers 3 GB of data over a network with an effective throughput of 120 Mbps. The Application Layer divides the data into requests of 3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

Answer:

Given

Parameter Value

Data size 3 GB = 3,000 MB

Request size 3 MB

Throughput 120 Mbps

Requests generated=1,000

$3,000 / 3 = 1,000$

Transmission time=200 s

Total bits = $3 \times 10^9 \times 8 = 24 \times 10^9$ bits

Time = $24 \times 10^9 / 120 \times 10^6 = 200$ s

Answer

- Requests = 1,000
- Transmission time $\approx$ 200 seconds

Application Layer Services

FTP provides file transfer commands, authentication, error reporting, and reliable user-level data exchange.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = 4 ms, Transport Layer = 3 ms, Network Layer = 5 ms, Data Link Layer = 2 ms, and Physical Layer = 1 ms. The propagation delay is 18 ms, and the transmission delay is 12 ms. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

Answer:

Delay Component Value

Application	4 ms
Transport	3 ms
Network	5 ms
Data Link	2 ms
Physical	1 ms
Propagation	18 ms
Transmission	12 ms

Total delay

$$4 + 3 + 5 + 2 + 1 + 18 + 12 = 45 \text{ ms}$$

Answer

Total end-to-end delay = 45 ms

Layer Contribution

- Application: user services
- Transport: segmentation and reliability
- Network: routing
- Data Link: framing and error checking
- Physical: signal transmission
- Propagation: travel time through the medium
- Transmission: time to place bits on the link

9.A network transfers 80 MB of useful data using frames of 1600 bytes, where each frame contains 80 bytes of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

Answer:

Given

Parameter	Value
Useful data	80 MB = 80,000,000 bytes
Frame size	1,600 bytes

Parameter	Value
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Overhead/frame	80 bytes
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Useful payload per frame

$$1,600 - 80 = 1,520 \text{ bytes}$$

Frames required

$$80,000,000 / 1,520 = 52,631.58 \rightarrow 52,632 \text{ frames}$$

Total overhead

$$52,632 \times 80 = 4,210,560 \text{ bytes}$$

Transmission efficiency

$$(80,000,000 / (80,000,000 + 4,210,560)) \times 100 \approx 95\%$$

Answer

- Frames required = 52,632
- Total overhead = 4,210,560 bytes
- Efficiency $\approx 95\%$

Effect of Overhead

Higher protocol overhead reduces useful bandwidth and increases transmission time.

10. A hospital transfers a patient's medical record of 240 MB through an IoT-enabled healthcare network. The Presentation Layer compresses the data by 30%, the Transport Layer divides the compressed data into 1400-byte segments, and the Data Link Layer adds 60 bytes of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Answer:**Given**

Parameter	Value
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Original file	240 MB
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Compression	30%
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Segment size	1,400 bytes
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Frame overhead	60 bytes
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Compressed file size=168 MB

$$240 \times 0.70 = 168 \text{ MB}$$

$$\text{Number of segments} = 120,000$$

$$168,000,000 / 1,400 = 120,000$$

$$\text{Protocol overhead} = 7,200,000 \text{ bytes}$$

$$120,000 \times 60 = 7,200,000 \text{ bytes} = 7.2 \text{ MB}$$

$$\text{Final transmitted data} = 175.2 \text{ MB}$$

$$168 \text{ MB} + 7.2 \text{ MB} = 175.2 \text{ MB}$$

Answer

Result	Value
Compressed size	168 MB
Segments	120,000
Overhead	7.2 MB

Final data transmitted 175.2 MB

How OSI Layers Work Together

- Presentation Layer: compresses and secures data.
- Transport Layer: segments data and ensures reliable delivery.
- Data Link Layer: adds framing and error detection.
- Physical Layer: transmits the final bit stream over the network.

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well structured answers with logical flow	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

Accuracy	10%	Completely accurate and covers all	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
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		required points			
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand